INCH-POUND
MIL-R-22684/3C
6 August 2001
SUPERSEDING
MIL-R-22684/3B
2 July 1970

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, INSULATED, STYLE RL32

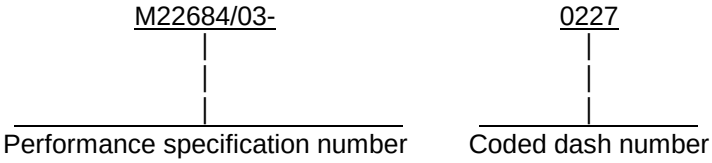
This specification is inactive for design after 2 July 1970. Use MIL-PRF-39017/3.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for insulated, film, fixed resistors of 2-percent and 5-percent resistance tolerance. These resistors are capable of full-load operation at an ambient temperature of 70°C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/°C).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:



The coded dash number is derived in accordance with MIL-PRF-22684.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DSCC-VAT, P.O. Box 3990 , Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the basic document or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-22684 - Resistors, Fixed, Film (Insulated), General Specification.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DoDSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related association specification, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-22684.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 1 watt based on full load operation at an ambient temperature of 70°C.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 500 volts.

3.5 Resistance values and resistance tolerances. The minimum and maximum standard resistance values and associated resistance tolerances shall be as listed in table I.

3.6 Terminal type. The terminal type available shall be in accordance with MIL-PRF-22684 and table I.

3.7 Dielectric withstanding voltage. Resistors shall be tested as specified in MIL-PRF-22684. The magnitude of test voltage shall be as follows:

Atmospheric pressure - 1,000 volts rms.
Barometric pressure - 500 volts rms.

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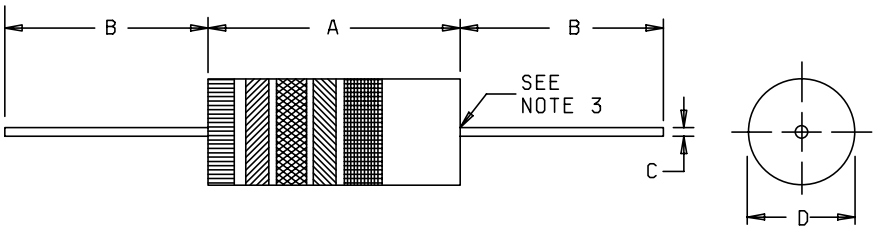

Ltr	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	0.520	0.593	13.21	15.06
B	1.375	1.625	34.92	41.28
C	.038	0.042	0.96	1.07
D	0.175	0.205	4.45	5.21

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18 mm) of the resistor body.

FIGURE 1. Configuration and dimensions.

3.8 Insulation resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the insulation resistance shall be not less than 100 megohms.

3.9 Moisture resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 1.5 percent.

3.10 Life. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 2.0 percent.

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Table I. PIN designation.

Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL32S100G	10	G	S	0001	RL32S121G	120	G	S	0053
RL32S100J	10	J	S	0002	RL32S121J	120	J	S	0054
RL32S110G	11	G	S	0003	RL32S131G	130	G	S	0055
RL32S110J	11	J	S	0004	RL32S131J	130	J	S	0056
RL32S120G	12	G	S	0005	RL32S151G	150	G	S	0057
RL32S120J	12	J	S	0006	RL32S151J	150	J	S	0058
RL32S130G	13	G	S	0007	RL32S161G	160	G	S	0059
RL32S130J	13	J	S	0008	RL32S161J	160	J	S	0060
RL32S150G	15	G	S	0009	RL32S181G	180	G	S	0061
RL32S150J	15	J	S	0010	RL32S181J	180	J	S	0062
RL32S160G	16	G	S	0011	RL32S201G	200	G	S	0063
RL32S160J	16	J	S	0012	RL32S201J	200	J	S	0064
RL32S180G	18	G	S	0013	RL32S221G	220	G	S	0065
RL32S180J	18	J	S	0014	RL32S221J	220	J	S	0066
RL32S200G	20	G	S	0015	RL32S241G	240	G	S	0067
RL32S200J	20	J	S	0016	RL32S241J	240	J	S	0068
RL32S220G	22	G	S	0017	RL32S271G	270	G	S	0069
RL32S220J	22	J	S	0018	RL32S271J	270	J	S	0070
RL32S240G	24	G	S	0019	RL32S301G	300	G	S	0071
RL32S240J	24	J	S	0020	RL32S301J	300	J	S	0072
RL32S270G	27	G	S	0021	RL32S331G	330	G	S	0073
RL32S270J	27	J	S	0022	RL32S331J	330	J	S	0074
RL32S300G	30	G	S	0023	RL32S361G	360	G	S	0075
RL32S300J	30	J	S	0024	RL32S361J	360	J	S	0076
RL32S330G	33	G	S	0025	RL32S391G	390	G	S	0077
RL32S330J	33	J	S	0026	RL32S391J	390	J	S	0078
RL32S360G	36	G	S	0027	RL32S431G	430	G	S	0079
RL32S360J	36	J	S	0028	RL32S431J	430	J	S	0080
RL32S390G	39	G	S	0029	RL32S471G	470	G	S	0081
RL32S390J	39	J	S	0030	RL32S471J	470	J	S	0082
RL32S430G	43	G	S	0031	RL32S511G	510	G	S	0083
RL32S430J	43	J	S	0032	RL32S511J	510	J	S	0084
RL32S470G	47	G	S	0033	RL32S561G	560	G	S	0085
RL32S470J	47	J	S	0034	RL32S561J	560	J	S	0086
RL32S510G	51	G	S	0035	RL32S621G	620	G	S	0087
RL32S510J	51	J	S	0036	RL32S621J	620	J	S	0088
RL32S560G	56	G	S	0037	RL32S681G	680	G	S	0089
RL32S560J	56	J	S	0038	RL32S681J	680	J	S	0090
RL32S620G	62	G	S	0039	RL32S751G	750	G	S	0091
RL32S620J	62	J	S	0040	RL32S751J	750	J	S	0092
RL32S680G	68	G	S	0041	RL32S821G	820	G	S	0093
RL32S680J	68	J	S	0042	RL32S821J	820	J	S	0094
RL32S750G	75	G	S	0043	RL32S911G	910	G	S	0095
RL32S750J	75	J	S	0044	RL32S911J	910	J	S	0096
RL32S820G	82	G	S	0045	RL32S102G	1,000	G	S	0097
RL32S820J	82	J	S	0046	RL32S102J	1,000	J	S	0098
RL32S910G	91	G	S	0047	RL32S112G	1,100	G	S	0099
RL32S910J	91	J	S	0048	RL32S112J	1,100	J	S	0100
RL32S101G	100	G	S	0049	RL32S122G	1,200	G	S	0101
RL32S101J	100	J	S	0050	RL32S122J	1,200	J	S	0102
RL32S111G	110	G	S	0051	RL32S132G	1,300	G	S	0103
RL32S111J	110	J	S	0052	RL32S132J	1,300	J	S	0104

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Table I. PIN designation - continued.

Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			0159
RL32S152G	1,500	G	S	0105	RL32S203G	20,000	G	S	0160
RL32S152J	1,500	J	S	0106	RL32S203J	20,000	J	S	0161
RL32S162G	1,600	G	S	0107	RL32S223G	22,000	G	S	0162
RL32S162J	1,600	J	S	0108	RL32S223J	22,000	J	S	
RL32S182G	1,800	G	S	0109	RL32S243G	24,000	G	S	0163
RL32S182J	1,800	J	S	0110	RL32S243J	24,000	J	S	0164
RL32S202G	2,000	G	S	0111	RL32S273G	27,000	G	S	0165
RL32S202J	2,000	J	S	0112	RL32S273J	27,000	J	S	0166
RL32S222G	2,200	G	S	0113	RL32S303G	30,000	G	S	0167
RL32S222J	2,200	J	S	0114	RL32S303J	30,000	J	S	0168
RL32S242G	2,400	G	S	0115	RL32S333G	33,000	G	S	0169
RL32S242J	2,400	J	S	0116	RL32S333J	33,000	J	S	0170
RL32S272G	2,700	G	S	0117	RL32S363G	36,000	G	S	0171
RL32S272J	2,700	J	S	0118	RL32S363J	36,000	J	S	0172
RL32S302G	3,000	G	S	0119	RL32S393G	39,000	G	S	0173
RL32S302J	3,000	J	S	0120	RL32S393J	39,000	J	S	0174
RL32S332G	3,300	G	S	0121	RL32S433G	43,000	G	S	0175
RL32S332J	3,300	J	S	0122	RL32S433J	43,000	J	S	0176
RL32S362G	3,600	G	S	0123	RL32S473G	47,000	G	S	0177
RL32S362J	3,600	J	S	0124	RL32S473J	47,000	J	S	0178
RL32S392G	3,900	G	S	0125	RL32S513G	51,000	G	S	0179
RL32S392J	3,900	J	S	0126	RL32S513J	51,000	J	S	0180
RL32S432G	4,300	G	S	0127	RL32S563G	56,000	G	S	0181
RL32S432J	4,300	J	S	0128	RL32S563J	56,000	J	S	0182
RL32S472G	4,700	G	S	0129	RL32S623G	62,000	G	S	0183
RL32S472J	4,700	J	S	0130	RL32S623J	62,000	J	S	0184
RL32S512G	5,100	G	S	0131	RL32S683G	68,000	G	S	0185
RL32S512J	5,100	J	S	0132	RL32S683J	68,000	J	S	0186
RL32S562G	5,600	G	S	0133	RL32S753G	75,000	G	S	0187
RL32S562J	5,600	J	S	0134	RL32S753J	75,000	J	S	0188
RL32S622G	6,200	G	S	0135	RL32S823G	82,000	G	S	0189
RL32S622J	6,200	J	S	0136	RL32S823J	82,000	J	S	0190
RL32S682G	6,800	G	S	0137	RL32S913G	91,000	G	S	0191
RL32S682J	6,800	J	S	0138	RL32S913J	91,000	J	S	0192
RL32S752G	7,500	G	S	0139					
RL32S752J	7,500	J	S	0140		Megohm			
RL32S822G	8,200	G	S	0141	RL32S104G	0.10	G	S	0193
RL32S822J	8,200	J	S	0142	RL32S104J	0.10	J	S	0194
RL32S912G	9,100	G	S	0143	RL32S114G	0.11	G	S	0195
RL32S912J	9,100	J	S	0144	RL32S114J	0.11	J	S	0196
RL32S103G	10,000	G	S	0145	RL32S124G	0.12	G	S	0197
RL32S103J	10,000	J	S	0146	RL32S124J	0.12	J	S	0198
RL32S113G	11,000	G	S	0147	RL32S134G	0.13	G	S	0199
RL32S113J	11,000	J	S	0148	RL32S134J	0.13	J	S	0200
RL32S123G	12,000	G	S	0149	RL32S154G	0.15	G	S	0201
RL32S123J	12,000	J	S	0150	RL32S154J	0.15	J	S	0202
RL32S133G	13,000	G	S	0151	RL32S164G	0.16	G	S	0203
RL32S133J	13,000	J	S	0152	RL32S164J	0.16	J	S	0204
RL32S153G	15,000	G	S	0153	RL32S184G	0.18	G	S	0205
RL32S153J	15,000	J	S	0154	RL32S184J	0.18	J	S	0206
RL32S163G	16,000	G	S	0155	RL32S204G	0.20	G	S	0207
RL32S163J	16,000	J	S	0156	RL32S204J	0.20	J	S	0208
RL32S183G	18,000	G	S	0157	RL32S224G	0.22	G	S	0209
RL32S183J	18,000	J	S	0158	RL32S224J	0.22	J	S	0210

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Table I. PIN designation - continued.

Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type Designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Megohm					Megohm			
RL32S244G	0.24	G	S	0211	RL32S514G	0.51	G	S	0227
RL32S244J	0.24	J	S	0212	RL32S514J	0.51	J	S	0228
RL32S274G	0.27	G	S	0213	RL32S564G	0.56	G	S	0229
RL32S274J	0.27	J	S	0214	RL32S564J	0.56	J	S	0230
RL32S304G	0.30	G	S	0215	RL32S624G	0.62	G	S	0231
RL32S304J	0.30	J	S	0216	RL32S624J	0.62	J	S	0232
RL32S334G	0.33	G	S	0217	RL32S684G	0.68	G	S	0233
RL32S334J	0.33	J	S	0218	RL32S684J	0.68	J	S	0234
RL32S364G	0.36	G	S	0219	RL32S754G	0.75	G	S	0235
RL32S364J	0.36	J	S	0220	RL32S754J	0.75	J	S	0236
RL32S394G	0.39	G	S	0221	RL32S824G	0.82	G	S	0237
RL32S394J	0.39	J	S	0222	RL32S824J	0.82	J	S	0238
RL32S434G	0.43	G	S	0223	RL32S914G	0.91	G	S	0239
RL32S434J	0.43	J	S	0224	RL32S914J	0.91	J	S	0240
RL32S474G	0.47	G	S	0225	RL32S105G	1.0	G	S	0241
RL32S474J	0.47	J	S	0226	RL32S105J	1.0	J	S	0242

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with Group A inspection and Group B inspection of MIL-PRF-22684 and as specified herein.

5. PACKAGING

5.1 Packaging. For acquisition purposes the packaging requirements shall be as specified in the contract or purchase order. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes The notes specified in MIL-PRF-22684 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, and complete PIN (see 1.2).

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements.

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6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, AT, AV, EA
Navy - CG, MC, OS
Air Force - 19
DLA - DH

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete block1s 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** **MIL-R-22684/3C**	**2. DOCUMENT DATE** (YYMMDD) 2001 August 6

3. DOCUMENT TITLE
RESISTORS, FIXED, FILM, INSULATED, STYLE RL32

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Incl Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus ATTN: DSCC – VAT	b. TELEPHONE (Include Area Code) (1) Commercial (614) 692-0553 (2) DSN 850-0553
c. ADDRESS (Include Zip Code) P.O. 3990 Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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